

Fig. 8: An actual-size, single-side PCB for relay section

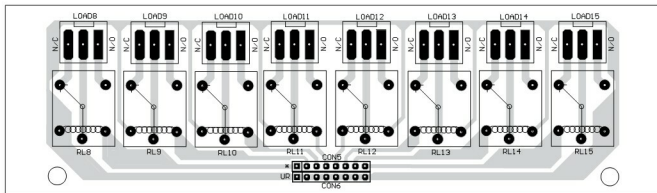


Fig. 9: Component layout for the PCB in Fig. 8

Power supply

The 230V AC mains is stepped down by transformer X1 to deliver a secondary output of 6V, 500 mA. The transformer output is rectified by a full-wave rectifier comprising diodes

D6 through D9, filtered by capacitor C8 and regulated by IC 7805 (IC8). Capacitor C7 bypasses the ripples present in the regulated supply. LED3 acts as the power indicator and R6 limits the current through LED3.

Software

Programs for the microcontrollers are written in BASIC and compiled using BASCOM Basic compiler for 8051 family. These are supported by Windows OS. The microcontrollers of the transmitter and receiver units are programmed with source programs 'Remote.bas' and 'Receiver.bas,' respectively. The bas codes are converted into hex codes using the above compiler. The hex codes are burnt into the respective microcontrollers using a suitable programmer.

Construction and testing

An actual-size, single-side PCB for the transmitter circuit (Fig. 2) is shown in Fig. 4 and its component layout in Fig. 5. The PCB for the receiver circuit excluding relay section (Fig. 3) is shown in Fig.

6 and its component layout in Fig. 7. The PCB for relays RL8 through RL15 connected to load 8 through load 15 is shown in Fig. 8 and its component layout in Fig. 9. You can use another PCB for relays RL1 through RL7 to connect load 1 through load 7. Suitable connectors are provided on the PCB. The receiver PCB is interfaced with the relay PCB by connecting CON1 to CON5, and CON2 to CON6.

After assembling the transmitter, receiver and relay sections on the respective PCBs, pull pin 1 of both HT12E and HT12D to ground. LED2 connected to V_p pin of the decoder should glow, indicating that a valid signal has been received. 255 sets of transmitter-receiver pairs can be used within the same area, each with a unique address. Alternately, we can also control 255 receivers with a remote control by changing the address. Now your RF-based multiple device control system is ready for use.

EFY note. The source codes of this project are available on www.efymag.com. ●

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Keys and Corresponding Device Activated

Keypad	Binary	Port pin	Device number
S1	0001	P1.1	1
S2	0010	P1.2	2
S3	0011	P1.3	3
S4	0100	P1.4	4
S5	0101	P1.5	5
S6	0110	P1.6	6
S7	0111	P1.7	7
S8	1000	P3.0	8
S9	1001	P3.1	9
S10	1010	P3.2	10
S11	1011	P3.3	11
S12	1100	P3.4	12
S13	1101	P3.5	13
S14	1110	P3.6	14
S15	1111	P3.7	15
S16	0000	P1.0	Reset all devices